

Cancer in South Florida Hispanic Women

A 9-Year Assessment

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ORIGINAL INVESTIGATION

Background: Cancer incidence varies among race and ethnic subgroups. Relatively little information exists on cancer among Hispanics, and none of it is stratified by race.

Methods: To explore and compare cancer incidence rates of each site of cancer among white Hispanic, white non-Hispanic, black Hispanic, and black non-Hispanic women in Dade County, South Florida, we accessed 9 years of cancer incidence data. Dade County's Hispanic population is the second largest of any US county. For each site, incidence rates, rate ratios, and 95% confidence intervals are presented.

Results: Among white Hispanics, there were statistically significant lower rates of cancers of the oral cavity, esophagus, colon, pancreas, lung, breast, ovary, bladder, kidney, melanoma, Kaposi's sarcoma, and non-

Hodgkin's lymphoma than among white non-Hispanics. Black Hispanics had significantly lower rates of cancers of the oral cavity, stomach, rectum, lung, cervix, and bladder than black non-Hispanics. However, white Hispanic women had significantly higher rates of cancers of the liver, gallbladder, and uterine cervix than white non-Hispanic women. No in situ breast cancer cases among blacks of either ethnic subgroup was found.

Conclusions: Both white Hispanic and black Hispanic women generally had lower rates of cancer than their race-specific non-Hispanic counterparts. However, incidence rates among Hispanics differ substantially by race; failure to distinguish them may lead to erroneous conclusions.

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FLORIDA HAS BEEN ONE OF THE three most rapidly growing states in the country for the last 40 years,¹ and is the fourth most populous state in the nation.² The Southeastern area, which includes Dade County, Florida, and its largest city, Miami, has consistently shown great changes in population, with an increase from 40 000 persons in 1920 to nearly 2 million in 1990. Dade County has become the county with the second largest Hispanic population in the United States, as well as the county with the second largest increase in Hispanics (Miami Herald, March 6, 1991). Dade County represents a substantial portion of the entire state's Hispanic community (60%). Dade County is best known as a center for Cuban immigration. Sixty percent of all Cubans in the United States live in Florida, 54% in Dade County alone. However, the proportion of Hispanics moving to Dade



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TABLE 1. Relative Risk of Cancer by Race, Ethnicity, and Site

Site	White	Black	Hispanic	Non-Hispanic
Overall	1.00	1.27	1.26	1.27
Stomach	1.00	0.39	0.85	0.85
Colon	1.00	0.44	0.31	0.31
Rectum	1.00	0.32	0.14	0.14
Esophagus	1.00	0.48	0.40	0.40
Liver	1.00	1.80	1.27	1.27
Gallbladder	1.00	1.97	1.43	1.43
Pancreas	1.00	0.71	0.61	0.61
Uterine	1.00	0.81	0.75	0.75
Cervix	1.00	0.81	0.75	0.75
Bladder	1.00	0.81	0.75	0.75
Prostate	1.00	0.81	0.75	0.75
Lung	1.00	0.81	0.75	0.75
Bronchus	1.00	0.81	0.75	0.75
Other	1.00	0.81	0.75	0.75

Among both white and black Hispanics, reports of cancers of the lung and bronchus were somewhat less frequent than for their non-Hispanic counterparts. In addition, among both Hispanic and non-Hispanic black women, and, to a lesser extent, among white Hispanic women, cancer of the uterine cervix was more frequent than among white non-Hispanic women. In contrast, among white Hispanic women, the rate for cancer of the uterine corpus ranked third, ie, higher than all other racial/ethnic groups. Multiple myeloma was also more frequently reported among both black Hispanic and non-Hispanic women.

In general, the rates among Hispanic women of either race were lower than those among the corresponding non-Hispanic subgroup. Significant differences in rates were observed for numerous cancer sites among whites; among blacks, fewer differences were observed. Among white women, cancers of the aerodigestive tract tended to be reported less frequently among Hispanics than among non-Hispanics. Among white Hispanic women, lower rates were observed for cancers of the oral cavity (rate ratio [RR]=0.48; 0.40 to 0.58), esophagus (RR=0.44; 0.31 to 0.64), colon (RR=0.81; 0.75 to 0.87), and pancreas (RR=0.71; 0.61 to 0.83). Cancers of both

See also pages 1057 and 1073

the liver and the gallbladder occurred at a significantly greater rate among white Hispanics than among white non-Hispanics (1.80; 1.27 to 2.54 and 1.97; 1.43 to 2.73), respectively. While RRs of cancer of the gallbladder and liver were not significantly elevated among black Hispanics women compared with black non-Hispanic women, the former had rates that were the highest of any race/ethnicity/gender combination, ie, more than twice as high as any other group.

No differences in the rates for cancers of the larynx, or of the nasal cavity and sinuses, were observed among white women. Among black Hispanic women, significantly lower rates for cancers of the oral cavity and pharynx (RR=0.35; 0.14 to 0.85), stomach (RR=0.39; 0.18 to 0.84), and rectum (RR=0.32; 0.14 to 0.73) were observed than among black non-Hispanic women.

SITE SPECIFIC ANALYSES

Table 1 presents aggregate data for all sites combined for each racial/ethnic subgroup considered in these analyses. The vast majority of all cases were diagnosed among white non-Hispanics and white Hispanics. After adjusting for the large discrepancies in the age distributions among the four racial/ethnic subgroups, although the standardized rates for all sites combined among Hispanics were substantially lower than those among non-Hispanics of each race, the differences in rates between racial groups was much greater among non-Hispanics than among Hispanics (ie, white vs black non-Hispanics). However, race has the same relative effect on risks among both Hispanic and non-Hispanics, and the rate ratio (white vs black) was 1.27 for Hispanics, and 1.26 for non-Hispanics.

ALL SITES COMBINED

RESULTS

Between non-Hispanics, this article addresses race-specific rates among South Florida Hispanics for the first time.

Dade County Hispanic women who were diagnosed with cancer during the period from 1981 through 1989 are included in this article. Analyses were generated from the Florida Cancer Data System, which collects information on malignant neoplasms diagnosed in all hospitals in Florida. The Florida Cancer Data System is a Surveillance, Epidemiology, and End Results' program of the National Cancer Institute comparative population-based registry, and has been independently estimated to be successful in capturing more than 95% of all cases diagnosed within the state.⁷ The ethnicity was regularly abstracted from the medical records. Reports essentially are based on self-classified ethnicity, ie, the method used by the Census Bureau. Cancers were coded according to the *International Classification of Diseases for Oncology*.⁸ Incidence rates were standardized to the US standard million population, to produce rates comparable with those published by other registries in the United States. Rates appearing in Table 1 and Table 2 are expressed in units of cases per 100 000 women-years. Race is presented as "black" or "white"; other races were excluded because of small numbers in Dade County. Denominators were based on interpolation of the 1980 and 1990 US censuses. Ethnicity is presented as "Hispanic" or "non-Hispanic." Age-standardized rate ratios of specific type/site of cancer associated with being Hispanic are presented with 95% two-sided confidence limits.

MATERIALS AND METHODS

Lower rates of the lung and bronchus cancers were observed among both Hispanic subgroups, regardless of racial group (RR=0.30; 0.27 to 0.32 and 0.37; 0.24 to 0.58, among whites and blacks, respectively). There were no significant differences observed for any other respiratory tract cancers, among either whites or blacks. Lower incidence rates among white Hispanic women were observed for cancers of the breast (RR=0.72; 0.69

to 0.75), ovary (RR=0.79; 0.70 to 0.89), and vulva (RR=0.56; 0.38 to 0.84) than among white non-Hispanic women, while significantly higher rates among white Hispanic women were observed for invasive cancers of the cervix (RR=1.21; 1.05 to 1.40). There were no significant differences in rates of cancers of the uterine corpus or vagina. Among black women, significantly lower rates for cervical cancer were observed among His-

*SRR indicates standardized rate ratio; CI, confidence interval. Numbers in parentheses indicate the number of cases of cancer.

Site	Rate	95% CI	Rate	95% CI	Rate	95% CI
Oral cavity, pharynx	7.8 (207)	3.8 (105)	0.50 (0.4-0.6)	5.5 (67)	1.8 (2)	0.25 (0.1-0.5)
Esophagus	1.7 (112)	0.7 (24)	0.44 (0.3-0.6)	0.4 (20)	4.5 (5)	0.07 (0.0-0.2)
Stomach	5.1 (372)	4.5 (207)	0.81 (0.6-1.1)	0.8 (62)	3.5 (3)	0.30 (0.2-0.5)
Small intestine	0.4 (25)	0.6 (24)	1.30 (0.6-2.5)	1.0 (7)	1.0 (7)	0.00 (0.0-0.0)
Colon	25.7 (2357)	25.5 (1219)	0.81 (0.6-0.9)	20 (271)	10.7 (14)	0.64 (0.4-1.0)
Rectum	10.4 (485)	8.8 (443)	0.95 (0.6-1.1)	7.5 (90)	2.4 (5)	0.52 (0.1-0.7)
Anus	1.7 (86)	1.6 (75)	0.95 (0.7-1.3)	0.7 (7)	0.7 (7)	0.00 (0.0-0.0)
Liver	1.1 (56)	2.0 (58)	1.80 (1.2-2.5)	1.7 (13)	2.3 (2)	1.50 (0.3-7.4)
Gallbladder	1.00 (61)	2.00 (66)	1.97 (1.4-2.7)	1.2 (13)	4.7 (4)	2.00 (0.3-24.0)
Pancreas	7.2 (617)	5.1 (230)	0.71 (0.5-0.8)	8.1 (63)	5.2 (4)	0.60 (0.2-1.5)
Nasal, sinus, and ear	0.5 (20)	0.4 (16)	0.72 (0.4-1.4)	0.7 (6)	0.7 (6)	0.00 (0.0-0.0)
Larynx	1.7 (82)	1.3 (56)	0.75 (0.5-1.1)	1.8 (21)	1.8 (21)	0.00 (0.0-0.0)
Lung and bronchus	43.9 (2502)	13.0 (587)	0.30 (0.3-0.3)	28.1 (273)	5.8 (6)	0.17 (0.2-0.6)
Mesothelioma	0.3 (15)	0.2 (9)	0.77 (0.3-1.8)	0.1 (1)	0.1 (1)	0.00 (0.0-0.0)
Heart, soft tissues	2.0 (94)	2.2 (80)	1.06 (0.6-1.5)	1.3 (18)	2.0 (2)	2.50 (0.2-18.4)
Bones, joints	0.7 (25)	0.7 (25)	1.06 (0.5-2.0)	0.6 (10)	2.0 (2)	1.10 (0.2-33.5)
Melanoma	6.3 (307)	1.9 (75)	0.29 (0.2-0.4)	0.9 (10)	0.9 (10)	0.00 (0.0-0.0)
Kaposi's sarcoma	0.4 (21)	0.1 (5)	0.36 (0.1-0.9)	0.3 (5)	0.3 (5)	0.00 (0.0-0.0)
Other nonmelanotic skin	0.5 (35)	0.3 (13)	0.54 (0.3-1.0)	0.3 (4)	0.3 (4)	0.00 (0.0-0.0)
Breast	100.9 (5201)	72.7 (3056)	0.72 (0.7-0.8)	70.7 (617)	65.5 (53)	0.64 (0.7-1.1)
Breast in situ	7.0 (307)	3.2 (130)	0.45 (0.4-0.5)	2.4 (28)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Cervix	8.8 (308)	10.7 (434)	1.21 (1.1-1.4)	18.4 (259)	13.0 (17)	0.62 (0.4-1.0)
Cervix in situ	18.4 (645)	9.0 (351)	0.49 (0.4-0.6)	16.5 (230)	11.9 (13)	0.72 (0.4-1.2)
Corpus	16.5 (1041)	17.1 (739)	0.91 (0.8-1.0)	15.0 (153)	13.4 (14)	0.90 (0.5-1.6)
Ovary	15.4 (788)	12.1 (489)	0.79 (0.7-0.9)	10.1 (150)	8.1 (7)	0.80 (0.4-1.6)
Vagina	0.7 (39)	0.6 (28)	0.86 (0.5-1.5)	1.4 (15)	0.7 (1)	0.10 (0.1-2.0)
Vulva	0.4 (37)	0.6 (37)	0.56 (0.4-0.8)	1.2 (1)	1.2 (1)	0.00 (0.1-3.0)
Bladder	7.8 (349)	4.9 (233)	0.64 (0.5-0.7)	4.1 (42)	1.2 (1)	0.28 (0.1-0.9)
Kidney	5.3 (320)	4.0 (162)	0.75 (0.6-0.9)	3.9 (42)	2.1 (2)	0.54 (0.2-1.7)
Eye and orbit	0.1 (5)	0.0 (1)	0.21 (0.3-1.5)	0.2 (2)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Brain, central nervous system	5.0 (217)	4.9 (170)	0.92 (0.7-1.2)	3.4 (47)	4.3 (3)	0.97 (0.3-3.2)
Thyroid	4.8 (180)	5.4 (212)	1.17 (1.0-1.4)	1.7 (25)	1.4 (2)	0.83 (0.2-3.2)
Hodgkin's lymphoma	2.0 (70)	1.9 (65)	0.96 (0.7-1.4)	0.8 (13)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Non-Hodgkin's lymphoma	9.7 (564)	7.4 (322)	0.76 (0.7-0.9)	0.7 (60)	2.8 (3)	0.54 (0.2-1.5)
Multiple myeloma	2.2 (150)	1.7 (75)	0.77 (0.6-1.0)	0.8 (61)	5.7 (6)	1.16 (0.3-2.8)
Acute lymphocytic leukemia	1.1 (26)	1.7 (43)	1.53 (0.9-2.5)	0.8 (8)	1.2 (1)	1.30 (0.1-20.3)
Chronic lymphocytic leukemia	1.5 (11)	0.7 (36)	0.49 (0.3-0.7)	0.9 (9)	1.6 (17)	2.38 (0.4-13.4)
Acute myeloid leukemia	2.5 (128)	1.9 (78)	0.74 (0.5-1.0)	1.6 (17)	3.7 (3)	2.38 (0.4-13.4)
Chronic myeloid leukemia	0.5 (41)	0.8 (33)	1.57 (0.9-2.7)	0.7 (9)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Other leukemias	0.6 (40)	0.4 (18)	0.67 (0.4-1.2)	0.5 (5)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Brain lymphoma	0.1 (2)	0.0 (1)	0.04 (0.0-0.5)	0.00 (0.0-0.0)	0.00 (0.0-0.0)	0.00 (0.0-0.0)
Other sites	19.3 (803)	16.3 (463)	0.86 (0.6-0.9)	15.5 (148)	10.2 (148)	0.74 (0.4-1.3)
All sites	354.9 (16 620)	248.7 (10 639)	0.70 (0.68-0.72)	281.8 (3031)	195.2 (175)	0.68 (0.6-0.8)

Table 2. Age Standardized Rate and RR of Cases of Cancer Among White Hispanic (WH), White Non-Hispanic (WNH), Black Hispanic (BH), and Black Non-Hispanic (BNH) Women*

Among other gynecologic sites, with the exception of ovarian cancer, there were no significant differences in the age-standardized incidence rates by ethnic subgroup. The lower rate ratio for ovarian cancer is consistent with ovarian cancer rates in Latin America, and also with increases among women who migrate to more developed trends.

Centers for Disease Control and Prevention data.¹⁶ Cervical cancer remains an important cause of morbidity and mortality in many Latin American countries,^{17,18} approaching that for breast cancer.¹⁹ In our analyses, the observation of lower breast cancer rates and higher cervical cancer rates among Hispanic subgroups, when compared with white non-Hispanics, generally reflects

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any race/ethnic subgroup. Although not statistically significant, white Hispanic women had the highest rate of thyroid cancer of any of the groups studied (including men). Thyroid cancers occurred 3.76 times as often among white Hispanic women as among black Hispanic women (1.85 to 7.65). Second, the only subgroup having higher rates of bone cancer than black Hispanic women were black Hispanic men. Finally, Kaposi's sarcoma was virtually nonexistent among either white or black Hispanic women.

Rates for most hematologic malignancies were similar among all racial/ethnic groups. However, among white Hispanic women, significantly lower rates were observed for non-Hodgkin's lymphoma (RR=0.79; 0.69 to 0.91), chronic lymphocytic leukemia (RR=0.49; 0.34 to 0.71), and acute non-lymphocytic leukemia (RR=0.74; 0.5 to 1.0), than among white non-Hispanic women. Despite the general lack of statistically significant rate ratios, black Hispanic women had the highest rate of multiple myeloma and acute non-lymphocytic leukemia of

The ratio of in situ/invasive breast and cervical cancer rates also varied by race and ethnicity (Figure 1). The ratio of in situ/invasive breast cancer rates among non-Hispanics (RR=0.62; 0.4 to 1.0),

In situ cancer by race and ethnicity: WHF, white Hispanic female; WHF, white Hispanic female; and WHF and BWHF, white and black Hispanic females, respectively.

and cancer of the rectum among South American born Hispanics were higher than those among the reference groups of non-Hispanic whites (none of these differences, however, attaining statistical significance). Similarly, Rosenwajke¹⁰ found the incidence of colorectal cancer to be lower among residents of Puerto Rico than among residents of the United States.

Model proposed by Pender,²⁹ it is important to determine the extent to which the different racial/ethnic groups have cancer related risk-increasing behavior that could explain these data. Race/ethnicity may act as, or interact with, a surrogate measure (such as genetic factors), or behave as effect modifiers (as for other environmental and cultural factors). A number of our findings that appear to implicate early environmental exposures are consistent not only with the results from migrant studies, but also with the multistage theory of carcinogenesis. That is, for those cancers whose incidence rates are altered after migrating from a country of lower to higher incidence (eg, breast and colorectal cancers), the data may relate to exposure to factors that act as promoters relatively late in the carcinogenic process (but not particularly to genetic factors). Conversely, for cancers whose incidence rates remain largely unaffected after migration (eg, cancer of the gallbladder), the rates may relate more to cultural exposures that act as initiators, at an earlier stage of carcinogenesis (or possibly to genetic factors). Further migrant studies could address the potential roles of such exposures as either initiators or promoters, focusing on the effects of respective early- and late-stage exposures.

These data also point to the need for distinguishing between Hispanic subgroups, not only by race, but by country of origin and acculturation. Representing diverse populations with a range of exposures by a single group will result in the averaging of exposures. As in our earlier article, which did not specify race, this averaging may inadvertently result in obscured associations. Furthermore, in South Florida, the black population is no more homogeneous than the Hispanic population, consisting of US born blacks, African born blacks, and Caribbean blacks; or the non-Hispanic white population—a mixture of native southerners, northern migrants, and western European immigrants. Thus, any associations found may be even more extreme than those presented here, while failing to find an association may reflect the inadequacies of the data, ie, absence of evidence is not evidence of absence. Thus, these data must be explored using focused studies having greater specificity to make etiologic inferences and/or to develop cancer control strategies.

Colorectal cancer is the second most frequently diagnosed cancer among women in the United States.⁸ While differences in rectal cancers were not observed among white Hispanic and non-Hispanic women, rates for rectal cancers among black Hispanic women were significantly lower than in black non-Hispanics. These findings are not inconsistent with those of Mack,²² who found lower rates of cancers of the colon and rectum among some Hispanic subgroups in Los Angeles, Calif. However, that study found that rates for cancer of the colon among Caribbean and South American born Hispanics are well for maintaining a relative dearth of smoking-related diseases.

The findings for liver and gallbladder cancers are consistent with our earlier analyses, and mirror the relationship seen in other American Hispanic populations.²² In addition to the likely role of hepatitis B virus in the cause of primary hepatocellular carcinoma,²² remaining cases are likely to be associated with hepatitis C virus, or with aflatoxin-B. A few studies^{24,25} suggest that Hispanics have higher prevalences of hepatitis B and C viruses. Aspergillus, which could thrive in the subtropical climate of South Florida, would only affect Hispanics preferentially, if they selectively consumed more of the foods likely to be infected.

Our earlier analysis did not detect a statistically significant elevation of gallbladder cancer for Hispanic women; this has changed with 9 years of data. Although cancer of the gallbladder accounts for fewer than 1% of all cancers in the United States, it is more common in Latin America, where it accounts for about 5% of all cancers among women.²⁶ Higher rates for cancers of the gallbladder were observed for Hispanic men in Miami, and among other Hispanic subgroups in the United States.^{27,28}

Incidence rates of lung cancer among Hispanic women of both racial groups were less than half and significantly lower than among their non-Hispanic counterparts. These findings are similar to those previously reported among Hispanics in South Florida, and among other subgroups of US Hispanic men and women (Hanson *Chronic News Service*, October 4, 1989). Nevertheless, large variations were observed in the male:female ratios, from less than 2.0 among white non-Hispanics, to nearly 5.0 among white Hispanics, and 5.0 among black Hispanics. For other smoking-related cancers, the results of our analyses did not consistently demonstrate significantly higher incidence rates among Hispanic women compared with non-Hispanics, in contrast to the significantly higher rates of laryngeal cancer among South Florida Hispanic men.

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